MTH 200 Chapter 5: Integration

5.2.3 Average Value Worksheet

1. Determine the average value of the function $f(x) = 10x + 3$ on the interval $[1, 3]$.

2. Determine the average value of the function $f(x) = |x| + 2$ on the interval $[-2, 4]$.

3. Find all numbers c that satisfy the MVT for Integrals for the function $f(x)=10x+3$ on the interval $[1,3]$.

4. Find all numbers c that satisfy the MVT for Integrals for the function $f(x)=|x|+2$ on the interval $[-2,4]$.

5. Suppose the function $k(x)$ is continuous on the interval $[1,8]$ and $\int_1^8 k(x)\,dx=28$. Show that the function $k(x)$ assumes the value of 4 at least once on the given interval.

6. Suppose that a car with whose velocity (in feet per second) after t seconds is given by $v(t) = 3t^2 + 4t - 1$. Show that the car must have been moving at a speed of 135 *ft/sec* at some point during the interval $[4,8]$